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Detailed mark scheme

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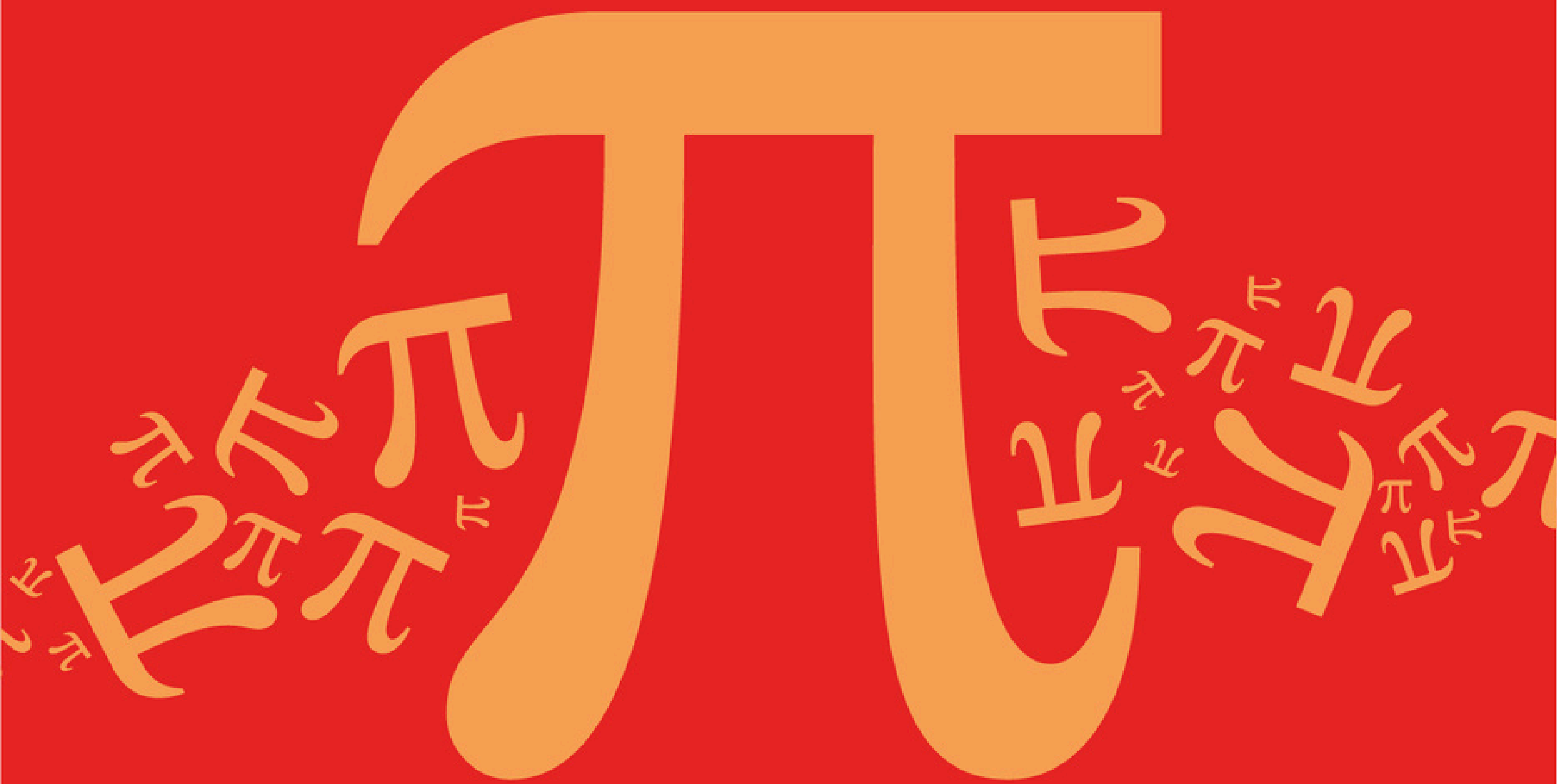
AQA A Level Further Mathematics (7367)



Topic **C: Matrices**

Sub topic **C5: Determinants (3x3 Determinants)**

Mark Scheme

AQA A Level Further Mathematics (7367)

Mark Scheme

Topic	**C: Matrices**
Sub topic	**C5: Determinants (3x3 Determinants)**

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses an appropriate method for finding the values of k (for example expanding appropriate determinant)	AO1.1a	M1	$\begin{vmatrix} 1 & -1 & k \\ k & -3 & 5 \\ 1 & -2 & 3 \end{vmatrix} = 0$
	Obtains a quadratic equation in k	AO1.1a	M1	$\begin{vmatrix} -3 & 5 \\ -2 & 3 \end{vmatrix} + \begin{vmatrix} k & 5 \\ 1 & 3 \end{vmatrix} + k\begin{vmatrix} k & -3 \\ 1 & -2 \end{vmatrix} = 0$
	Obtains two correct values for k	AO1.1b	A1	$1 + 3k - 5 + k(-2k + 3) = 0$ $-2k^2 + 6k - 4 = 0$ $k^2 - 3k + 2 = 0$ $(k - 2)(k - 1) = 0$ $k = 2$ or 1
(b)	Selects an appropriate method to determine the appropriate geometrical configuration and substitutes ‘their’ first value of k	AO3.1a	M1	when $k = 1$ $x - y + z = 3$ $x - 3y + 5z = -1$ $x - 2y + 3z = -4$
	Eliminates one variable or uses row reduction	AO1.1a	M1	$-2y + 4z = -4$ $y - 2z = 7$
	Obtains a contradiction and makes correct deduction about the geometric configuration (must have correct value for k)	AO2.2a	R1	$y - 2z = 2$; $y - 2z = 7$ Hence equations are inconsistent and the three planes form a prism
	Substitutes ‘their’ 2nd value of k into selected method to determine the appropriate geometrical configuration	AO1.1a	M1	when $k = 2$ $x - y + 2z = 3$ $2x - 3y + 5z = -1$ $x - 2y + 3z = -4$
	Obtains a consistent set of equations and makes correct deduction about geometric configuration	AO2.2a	R1	$R_2 - 2R_1 : -y + z = -7$ $R_3 - R_1 : -y + z = -7$

	(must have correct value for k)			Hence equations are consistent and the three planes form a sheaf − they meet in line
(c)	Deduces that the planes must meet in a line and hence that $k = 2$	AO2.2a	R1	$x - y + 2z = 3$ $2x - 3y + 5z = -1$ $x - 2y + 3z = -4$ $\Rightarrow -y + z = -7$ Let $z = \lambda$ Then $y = \lambda + 7$ and $x = 3 + y - 2z$ $= 3 + \lambda + 7 - 2\lambda$ $= -\lambda + 10$
	Selects method to find solution: For example, sets one variable = λ, substitutes and attempts to find other variables in terms of λ	AO1.1a	M1	**ALT** $\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \times \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$
	Fully states correct solution CAO	AO1.1b	A1	$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 10 \\ 7 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$
				Total 11 marks

Q2.

(a) det $\mathbf{A}^{-1} = -3 \Rightarrow$ det $\mathbf{A} = -\frac{1}{3}$

B1

1

(b) det($\mathbf{AB}$) = 24 $\Rightarrow$ det $\mathbf{B} = \frac{24}{\det \mathbf{A}} = -72$

*M1 for use of det(**AB**) = det **A** × det **B***

M1

*A1F ft their det**A***

A1F

Volume = 20 × 72 = 1440 cm^3

Must be positive

A1cso

3

[4]

Q3.

(a) $\left[\begin{array}{ccc|c} 2 & -1 & -1 & 3 \\ 1 & 2 & -3 & 4 \\ 2 & 1 & a & b \end{array}\right]$

$r_3 \to r_3 - r_1$
$r_2 \to 2r_2 - r_1$ $\left[\begin{array}{ccc|c} 2 & -1 & -1 & 3 \\ 0 & 5 & -5 & 5 \\ 0 & 2 & a+1 & b-3 \end{array}\right]$

Correct row operations used to create two zeros in first column − coefficients must be correct

M1

$r_3 \to 5r_3 - 2r_2$ $\left[\begin{array}{ccc|c} 2 & -1 & -1 & 3 \\ 0 & 5 & -5 & 5 \\ 0 & 0 & 5a+15 & 5b-25 \end{array}\right]$

Use of row operations to create third zero in second column or compare ratios of coefficients in rows 2 and 3

M1

No unique solution: $5a + 15 = 0$

$a = -3$

Solves equation with coefficient of z = 0 or equation formed from comparison of ratios (eg $a + 1 = -2$) $a = -3$ is a printed answer

A1

3

Alternative 1:

$2x - y - z = 3$ ①
$x + 2y - 3z = 4$ ②
$2x + y + a z = b$ ③

① + ③ ⇒ $4x + (a - 1) z = b + 3$ ④
2③ − ② ⇒ $3x + (2a + 3) z = 2b - 4$ ⑤

Correct elimination of 1 variable.
Coefficients must be correct.

(M1)

$4⑤ - 3④ \Rightarrow (5a + 15)\,z = 5b - 25$

Correctly reduce to one equation with a, b

(M1)

$5a + 15 = 0$
$a = -3$

Solves equation with coefficient of z = 0

(A1)
(3)

Alternative 2:

Solve $\begin{vmatrix} 2 & -1 & -1 \\ 1 & 2 & -3 \\ 2 & 1 & a \end{vmatrix} = 0$

$2\begin{vmatrix} 2 & -3 \\ 1 & a \end{vmatrix} - 1\begin{vmatrix} -1 & -1 \\ 1 & a \end{vmatrix} + 2\begin{vmatrix} -1 & -1 \\ 2 & -3 \end{vmatrix} = 0$

Correct expansion by row or column

(M1)

$2(2a + 3) - (-a + 1) + 2\,(5) = 0$

Correct expansion of 2 by 2 determinants

(M1)

$5a + 15 = 0$
$a = -3$

Solves equation with determinant = 0

(A1)
(3)

(b) Either $5b - 25 \neq 0$ or $5b - 25 = 0$

Sets their constant ≠ 0 (or 0)

M1

Inconsistent $b \neq 5$

CSO (accept $b > 5$, b

A1
2
[5]

Q4.

(a) First factor (quadratic) = $x^2 + y^2 + z^2$

Quadratic factor identified anywhere

$$\left(x^2+y^2+z^2\right)\begin{vmatrix} x^2-x & y^2-y & z^2-z \\ x & y & z \\ 1 & 1 & 1 \end{vmatrix}$$

B1

$C_2 \to C_2 - C_1$

Correct use of a column/row operation to obtain first linear factor − no more than one slip

M1

$C_3 \to C_3 - C_1$

$(x^2 + y^2 + z^2)$

$$\begin{vmatrix} x^2-x & y^2-x^2-(y-x) & z^2-x^2-(z-x) \\ x & y-x & z-x \\ 1 & 0 & 0 \end{vmatrix}$$

Two linear factors $(y - x)$ and $(z - x)$

$$\left(x^2+y^2+z^2\right)(y-x)(z-x)\begin{vmatrix} x^2-x & y+x-1 & z+x-1 \\ x & 1 & 1 \\ 1 & 0 & 0 \end{vmatrix}$$

Two correct linear factors found $(y - x)(z - x)$ or equivalent

A1,A1

Expand to get $(x^2 + y^2 + z^2)(y - x)(z - x)(y - z)$

Finding the final linear factor $(y - z)$ or equivalent

M1

All correct - CSO

A1

6

(b) x, y, z distinct $\Rightarrow x \neq y \neq z$

$\Rightarrow (y - x)(z - x)(y - z) \neq 0$

Explaining why linear factors are ≠ 0

E1

x, y, z distinct, real $\Rightarrow x^2 + y^2 + z^2 \neq 0$

$\Rightarrow \Delta \neq 0$

Explaining why the quadratic factor is ≠ 0

E1

2

[8]

Q5.

(a) $$\begin{vmatrix} 1-\lambda & 1 & 2 \\ 0 & 2-\lambda & 2 \\ -1 & 1 & 3-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)\begin{vmatrix} 2-\lambda & 2 \\ 1 & 3-\lambda \end{vmatrix} - \begin{vmatrix} 1 & 2 \\ 2-\lambda & 2 \end{vmatrix} = 0$$

Correct row / column expansion of $|\mathbf{M} - \lambda\mathbf{I}| = 0$

M1

$(1 - \lambda)(\lambda - 4)(\lambda - 1) - 2(\lambda - 1) = 0$
$(1 - \lambda)\ [(2 - \lambda)(3 - \lambda) - 2] - [2 - 2(2 - \lambda)] = 0$

Correct expansion of 2 by 2 determinants
− dependent on first M1

m1

$[\lambda - 1][-\lambda^2 + 5\lambda - 6] = 0$

or $-\lambda^3 + 6\lambda^2 - 11\lambda + 6 = 0$

A1

$-(\lambda - 1)(\lambda - 2)(\lambda - 3) = 0$

Attempt to show $\lambda = 2$ is an eigenvalue

M1

$\lambda = 1, 2$ or 3

Fully correct eigenvalues - CAO

A1

5

(b) $$\begin{bmatrix} -1 & 1 & 2 \\ 0 & 0 & 2 \\ -1 & 1 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Attempt to solve $(\mathbf{M} - 2\mathbf{I})\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

M1

$\Rightarrow 2z = 0$ $\quad\Rightarrow z = 0$

$-x + y + z = 0$ $\quad x = y$

Both relationships obtained (can be unsimplified)

A1

$\Rightarrow \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ is an eigenvector

Stated clearly; accept multiples

A1 **3**

(c) (i) $\begin{pmatrix} 2 & 2 & -3 \\ 2 & 2 & 3 \\ -3 & 3 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \\ 0 \end{pmatrix} = 4\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

Attempt at $\mathbf{N}\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

M1

$\Rightarrow \lambda = 4$ is an eigenvalue

A1 **2**

(ii) Eigenvector = $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

Accept multiples − part (b) must be correct

B1

Eigenvalue = 2 × 4 = 8

M1 for product of relevant eigenvalues

M1A1 **3**

Alternative:

Eigenvector = $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

Accept multiples − part b) must be correct

(B1)

$$\begin{pmatrix} 1 & 1 & 2 \\ 0 & 2 & 2 \\ -1 & 1 & 3 \end{pmatrix}\begin{pmatrix} 2 & 2 & -3 \\ 2 & 2 & 3 \\ -3 & 3 & 3 \end{pmatrix} = \begin{pmatrix} -2 & 10 & 6 \\ -2 & 10 & 12 \\ -9 & 9 & 15 \end{pmatrix}$$

$$\begin{pmatrix} -2 & 10 & 6 \\ -2 & 10 & 12 \\ -9 & 9 & 15 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 8 \\ 8 \\ 0 \end{pmatrix} = 8\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

Multiplies to get MN and attempts to find eigenvalue

(M1)

So eigenvalue is 8

Correct eigenvalue

(A1)

(3)

[13]

Q6.

(a) Det (AB) = (4)(1) − (1)(1) = 3 ≠ 0

Must state non-zero or ≠ 0

B1

1

(b) Matrix of cofactors $\begin{bmatrix} 0 & 0 & 3 \\ 4 & -1 & 8-4k \\ -1 & 1 & k-8 \end{bmatrix}$

Attempt at matrix of cofactors − at least five correct entries

M1

Fully correct matrix of cofactors

A1

$(AB)^{-1} = \frac{1}{3}\begin{bmatrix} 0 & 4 & -1 \\ 0 & -1 & 1 \\ 3 & 8-4k & k-8 \end{bmatrix}$

Their cofactor matrix transposed correctly

M1

At least five correct = A1 (exclude effect of determinant)
All entries fully correct = A2

A2,1

5

(c) $(AB)^{-1} = B^{-1}A^{-1}$

$\Rightarrow B^{-1} = (AB)^{-1} A$

$B^{-1} = \frac{1}{3}\begin{bmatrix} 0 & 4 & -1 \\ 0 & -1 & 1 \\ 3 & 8-4k & k-8 \end{bmatrix}\begin{bmatrix} k & 6 & 8 \\ 0 & 1 & 2 \\ -3 & 4 & 8 \end{bmatrix}$

Use of $(AB)^{-1}$ and A multiplied

M1

Correct order of multiplication

A1

$= \frac{1}{3}\begin{bmatrix} 3 & 0 & 0 \\ -3 & 3 & 6 \\ 24 & -6 & -24 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 2 \\ 8 & -2 & -8 \end{bmatrix}$

All correct = A2
5+ correct = A1 (exclude effect of determinant)

NB − if an attempt is made to find B by setting up a system of simultaneous equations then

M1 − 9 correct equations used

A1 for $B = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 2 & \frac{1}{2} \\ \frac{3}{2} & -\frac{1}{2} & -\frac{1}{4} \end{pmatrix}$

Final A2 as above

A2,1

4

[10]

Q7.

(a)

$\begin{vmatrix} 2 & n & 1 \\ 3 & -1 & n \\ -1 & 7 & 1 \end{vmatrix}$

Expanding the det. of the coefft. mtx.

M1

Setting it = 0

M1

Obtaining & solving a quadratic eqn. in n

M1

$0 = n^2 + 17n - 18 = (n + 18)(n - 1) \Rightarrow n = 1, -18$

CSO

A1

4

(b) $n = 1$ gives $2x + y + z = 5$
$3x - y + z = 1$
$-x + 7y + z = 1$
ft their chosen integer n

B1

Eliminating one variable from a pair of equations, twice

M1

e.g. ② − ① ⇒ $x - 2y = -y = -4$

A1 ft

and ② − ③ ⇒ $4x - 8y = 0$

A1 ft

Inconsistency clearly demonstrated from fully correct working

E1

3 planes have no common intersection (or form a Δʳ prism)
Also ft "3 planes meet in a common line" or
"3 planes form a sheaf" if consistency conclusion made

B1 ft

6

[10]

Q8.

(a) eg $\begin{vmatrix} yz & x(y+z) & xy \\ x & y+z & z \\ x^2 & z^2-y^2 & z^2 \end{vmatrix}$

$C_2' = C_2 + C_3$

M1

$= (y+z)\begin{vmatrix} yz & x & xy \\ x & 1 & z \\ x^2 & z-y & z^2 \end{vmatrix}$

A1

2

(b) eg $(y+z)\begin{vmatrix} y(z-x) & x & xy \\ x-z & 1 & z \\ x^2-z^2 & z-y & z^2 \end{vmatrix}$

$C_1' = C_2 - C_3$

Attempt at a second linear factor

M1

$$= (y - z)(y + z)\begin{vmatrix} -y & x & xy \\ 1 & 1 & z \\ x+z & z-y & z^2 \end{vmatrix}$$

A1

$$= (y - z)(y + z)\begin{vmatrix} -(x+y) & x & xy \\ 0 & 1 & z \\ x+y & z-y & z^2 \end{vmatrix}$$

$C_1' = C_2 + C_2$

Complete attempt at remaining factors (M0 if they just expand and can do nothing with it)

M1

$= (x - z)(y + z)(x + y)(xz - xy - yz)$

A1

4

[6]

Q9.

(a) (i) $\mathbf{A}^T = \begin{bmatrix} k & 2 & 1 \\ 1 & k & 2 \\ 2 & 1 & k \end{bmatrix}$

B1

Good multiplication attempt at **A A**^{T}

*If they multiply **A**^{T} **A** instead, they can score B1 M1 A0 A0 A1 A1*

A1A1

M1

$$= \begin{bmatrix} k^2+5 & 3k+2 & 3k+2 \\ 3k+2 & k^2+5 & 3k+2 \\ 3k+2 & 3k+2 & k^2+5 \end{bmatrix}$$

Main diagonal correct

A1

All others correct

A1

$k = -\frac{2}{3}$ $\quad m = 5\frac{4}{9}$ or $\frac{49}{9}$

6

(ii) $\begin{bmatrix} -\frac{2}{3} & 1 & 2 \\ 2 & -\frac{2}{3} & 1 \\ 1 & 2 & -\frac{2}{3} \end{bmatrix}^{-1} = \frac{9}{49}\begin{bmatrix} -\frac{2}{3} & 2 & 1 \\ 1 & -\frac{2}{3} & 2 \\ 2 & 1 & -\frac{2}{3} \end{bmatrix}$

ft $\frac{1}{\text{their } m}$ *and their* k

B1

Accept $\frac{9}{49}\begin{bmatrix} k & 2 & 1 \\ 1 & k & 2 \\ 2 & 1 & k \end{bmatrix}$ since the value of k is now known,

but not just $\frac{9}{49}$ $\mathbf{A}^T$

Decimal version (correct to at least 3sf) is also ok:

$\begin{bmatrix} -0.122 & 0.367 & 0.184 \\ 0.184 & -0.122 & 0.367 \\ 0.367 & 0.184 & -0.122 \end{bmatrix}$

1

(b) (i) det $\mathbf{A}$ = $k^3 - 6k + 9$

Good attempt (cubic)

M1
A1

2

(ii) det $\mathbf{A}$ = $(k + 3)(k^2 - 3k + 3)$

Factorisation attempt

M1

$k = -3$

A1

$\Delta = 9 - 12$

B1

Special Cases

k = − 3 but no working

(B1)

For all 3 roots, − 3 and $\frac{3 + i\sqrt{3}}{2}$ *given with no supporting working*

(B3)

3

(iii) Replacing k by $(k - 7)$

NOT just in the determinant form (ie starting again)

M1

$k = 4$ (however obtained)

ft their previous $k + 7$

A1

2

[14]

Q10.

(a) $\Delta = \begin{vmatrix} 1 & 2 & 3 \\ x & y & z \\ x+y+z & y+z+x & z+x+y \end{vmatrix}$

e.g.

$R_3' = R_3 + R_2$

M1

$= (x+y+z)\begin{vmatrix} 1 & 2 & 3 \\ x & y & z \\ 1 & 1 & 1 \end{vmatrix}$

A1

2

(b) Expanding remaining det.

M1

$\Delta = (x + y + z)(x - 2y + z)$

A1

2

[4]

Q11.

(a) det **A** = $5p - 1$

B1

det **B** = $p^2 - 10p - 11$

M1A0 if num error(s) made

M1A1

3

(b) Use of det(**AB**) = det **A** det **B**

PI

B1

Finding three values of p

Allow correct factors here

M1

$p = \frac{1}{5}$, 11, –1

ft numerical errors in (a)

A1F

3

[6]

Q12.

(a) (i) Appropriate row/column operation

eg $R_1' = R_1 + R_3$, $R_3' = R_3 + R_1$

or $C_3' = C_3 - nC_2$

M1

$$\Delta = \begin{vmatrix} n^2+n+1 & 0 & 0 \\ 0 & 1 & n \\ 1 & -(n+1) & 1 \end{vmatrix}$$

$$\ldots = (n^2+n+1)\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & n \\ 1 & -(n+1) & 1 \end{vmatrix}$$

Factor correctly extracted

A1

2

(ii) Expanding remaining determinant

OE

M1

$\Delta = (n^2 + n + 1)^2$

A1

2

(b) $\Delta = (n^2 + n)^2 + 2n^2 + 2n + 1$

Accept unsimplified

B1

$\ldots = (n^2 + n)^2 + (n + 1)^2 + n^2$

B1

2

(c) Setting $n = 10$

M1

$111^2 = 12321 = 110^2 + 11^2 + 10^2$

A1

2

[8]

Q13.

(a) (i) $R_1' = R_1 + R_2$ or $R_2' = R_2 + R_1$

M1

leading to $R_{1/2} = (4 - q \quad 4 - q \quad 0)$

A1

2

(ii)

$$\Delta = (4-q)\begin{vmatrix} 1 & 1 & 0 \\ -12 & -1-q & -7 \\ 6 & 6 & 10-q \end{vmatrix}$$

$$= (4-q)\begin{vmatrix} 0 & 1 & 0 \\ q-11 & -1-q & -7 \\ 0 & 6 & 10-q \end{vmatrix}$$

By $C_1' = C_1 - C_2$ (eg)

M1

$$= (4-q)(q-11)\begin{vmatrix} 0 & 1 & 0 \\ 1 & -1-q & -7 \\ 0 & 6 & 10-q \end{vmatrix}$$

2^{nd} linear factor correct

A1

$= (4 - q)(q - 11) \times \ldots$

Full attempt at 3^{rd} factor

M1

$= (4 - q)(q - 11)(q - 10)$

A1

4

(b) (i)

$$\begin{bmatrix} 16 & 5 & 7 \\ -12 & -1 & -7 \\ 6 & 6 & 10 \end{bmatrix}\begin{bmatrix} 2 \\ 5 \\ -7 \end{bmatrix} = \begin{bmatrix} 8 \\ 20 \\ -28 \end{bmatrix}$$

M1A1

$= 4\begin{bmatrix} 2 \\ 5 \\ -7 \end{bmatrix}$ so that $\lambda = 4$

A1

3

(ii) $\lambda = 10, 11$ noted or used

ft

B1

$$\lambda = 10 \Rightarrow \begin{aligned} 0 &= 6x + 5y + 7z \\ 0 &= -12x - 11y - 7z \\ 0 &= 6x + 6y \end{aligned}$$

B1

Using $y = -x$ to get $z \sim x/y$

M1

Eigenvector (–7, 7, 1)

Any non-zero multiple will do

A1

$$\lambda = 11 \Rightarrow \begin{aligned} 0 &= 5x + 5y + 7z \\ 0 &= -12x - 12y - 7z \\ 0 &= 6x + 6y - z \end{aligned}$$

Either $y = -x$ or $z = 0$

M1

Eigenvector (1, –1, 0)

Any non-zero multiple will do

A1

7

(c) $\frac{x}{2} = \frac{y}{5} = \frac{z}{-7}$, $\frac{x}{-7} = \frac{y}{7} = z$

Any line eqns.

M1

or $x = -y$, $z = 0$

any one ft correct

A1

2

[18]

Q14.

$$\Delta = \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ yz & zx & xy \end{vmatrix}$$

$$= \begin{vmatrix} x & y-x & z-x \\ x^2 & y^2-x^2 & z^2-x^2 \\ yz & z(x-y) & y(x-z) \end{vmatrix}$$

By $C_2' = C_2 - C_1$ *(eg)*

M1

$C_3' = C_3 - C_1$ *(eg)*

M1

$$= (y-x)(z-x)\begin{vmatrix} x & 1 & 1 \\ x^2 & y+x & z+x \\ yz & -z & -y \end{vmatrix}$$

First two factors extracted (what's left has to be correct also)

A1A1

$$= (y-x)(z-x)\begin{vmatrix} x & 1 & 0 \\ x^2 & y+x & z-y \\ yz & -z & z-y \end{vmatrix}$$

By $C_3' = C_3 - C_2$ *(e.g.)*

M1

$$= (y-x)(z-x)(z-y)\begin{vmatrix} x & 1 & 0 \\ x^2 & y+x & 1 \\ yz & -z & 1 \end{vmatrix}$$

3rd factor extracted

A1

$= (x - y)(y - z)(z - x)(xy + yz + zx)$

Further R/C ops or expansion of remaining det (almost a dM1)

M1

CAO up to equivalents due to re-positioning of the signs

A1

Alternatives using *Cyclic Symmetry* and the *Factor Theorem* are fine

[8]

Q15.

(a) $D = x^2 + y^2 + z^2 - xy - yz - zx$

M1A1

2

(b) E.g. by $C_1' = C_1 + (C_2 + C_3)$

M1

$$\Rightarrow \Delta = \begin{vmatrix} x+y+z & y & z \\ 0 & z-x & x-y \\ 2(x+y+z) & y+x & z+y \end{vmatrix}$$

$$= (x+y+z)\begin{vmatrix} 1 & y & z \\ 0 & z-x & x-y \\ 2 & y+x & z+y \end{vmatrix}$$

Shown or explained from previous line

A1

2

(c) Working on (R/C-ops) or expanding

M1

remaining determinant
2^{nd} factor = $-(x^2 + y^2 + z^2 - xy - yz - zx)$
Good attempt

dM1

$k = -1$

A1

3

[7]

Q16.

(a) Det(**M**) = $a^3 + b^3 + c^3 - 3abc$
Good attempt; correct

M1A1

2

(b) $\begin{bmatrix} ad+bf+ce & ae+bd+cf & af+be+cd \\ af+be+cd & ad+bf+ce & ae+bd+cf \\ ae+bd+cf & af+be+cd & ad+bf+ce \end{bmatrix}$

M1

At least 5 correct;

A1

all 9 correct

A1

3

(c) Use of det(**MN**) = det(**M**) det(**N**)

M1

$x = ad + bf + ce$, $y = ae + bd + cf$ and
$z = af + be + cd$

All correctly identified
Give B1 (SC) if just this with no explanation why

A1

2

[7]

Q17.

(a) eg 3 × (1) – (2) $\Rightarrow$ 13y + 13z = –13

Eliminating first variable

M1

(3) – (2) $\Rightarrow$ 15y + 11z = –5

A1A1

Solving 2 × 2 system

M1

$x = 6$, $y = 1½$, $z = –2½$

A1

5

Alt I (Cramer's Rule):

$\Delta = \begin{vmatrix} 1 & 3 & 5 \\ 3 & -4 & 2 \\ 3 & 11 & 13 \end{vmatrix}$, $\Delta_x = \begin{vmatrix} -2 & 3 & 5 \\ 7 & -4 & 2 \\ 2 & 11 & 13 \end{vmatrix}$,

$$\Delta_y = \begin{vmatrix} 1 & -2 & 5 \\ 3 & 7 & 2 \\ 3 & 2 & 13 \end{vmatrix}, \ \Delta_z = \begin{vmatrix} 1 & 3 & -2 \\ 3 & -4 & 7 \\ 3 & 11 & 2 \end{vmatrix}$$

Attempt at any two

(M1)

= 52, 312, 78 and – 130 respectively

Δ correct; ≥1 other determinant correct

(A1A1)

$$x = \frac{\Delta_x}{\Delta}, \ y = \frac{\Delta_y}{\Delta}, \ z = \frac{\Delta_z}{\Delta}$$

At least one attempted numerically

(M1)

$x = 6$, $y = 1½$, $z = -2½$

(A1)

(5)

Alt II (Augmented matrix method):

$$\left[\begin{array}{ccc|c} 1 & 3 & 5 & -2 \\ 3 & -4 & 2 & 7 \\ 3 & 11 & 13 & 2 \end{array}\right] \rightarrow$$

(M1)

$$\left[\begin{array}{ccc|c} 1 & 3 & 5 & -2 \\ 0 & -13 & -13 & 13 \\ 0 & 2 & -2 & 8 \end{array}\right]$$

$R_2 \rightarrow R_2 - 3R_1$
$R_3 \rightarrow R_3 - 3R_1$

(A1)

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 3 & 5 & -2 \\ 0 & 1 & 1 & -1 \\ 0 & 1 & -1 & 4 \end{array}\right]$$

(A1)

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 3 & 5 & -2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & -2 & 5 \end{array}\right]$$

$R_3 \rightarrow R_3 - R_2$

Substituting back to get
$x = 6$, $y = 1½$, $z = –2½$

(M1A1)

(5)

Alt III (Inverse matrix method):

$$C^{-1} = \frac{1}{52}\begin{bmatrix} -74 & 16 & 26 \\ -33 & -2 & 13 \\ 45 & -2 & -13 \end{bmatrix}$$

M0 if no inverse matrix is given

(M1A1A1)

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = C^{-1}\begin{bmatrix} -2 \\ 7 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 1.5 \\ -2.5 \end{bmatrix}$$

(M1A1)

(5)

(b) (i) $$\begin{vmatrix} 1 & 3 & 5 \\ 3 & -4 & 2 \\ a & 11 & 13 \end{vmatrix} = 26a - 26$$

Attempt at determinant; OE

M1

Setting equal to zero and solving for a

m1

$a = 1$

A1

3

(ii) $x + 3y + 5z = -2$
$3x - 4y + 2z = 7$
$x + 11y + 13z = b$

NB $y + z = -1$ (from before)

B1

(3) – (1) $\Rightarrow$ $8y + 8z = b + 2$

B1

$b + 2 = -8 \Rightarrow b = -10$

Equating; CAO

M1A1

4

Alternative:
Substituting $x = 6$, $y = 1½$, $z = –2½$
into $x + 11y + 13z = b$

(M3)

$\Rightarrow b = –10$

Since, to be consistent, the 3rd plane must contain the line of intersection of the first 2 planes, and therefore contains this point

(A1)

(4)

[12]

Q18.

(a) Det **A** = $k + 3 + 12 – 4 – 9 – k = 2$

M1

CAO

A1

2

(b) $\mathbf{A}^{-1} = \dfrac{1}{\text{Det A}}(\text{adj } \mathbf{A})$

Correct use of the determinant (any value)

B1

Attempt at matrix of cofactors

M1

$$= \frac{1}{2}\begin{bmatrix} k-9 & 3-k & 2 \\ 12-k & k-4 & -2 \\ -1 & 1 & 0 \end{bmatrix}$$

*Use of transposition **and** signs*

M1

At least 5 entries correct (even if 2nd M1 not earned)

A1

CAO – ft det only

A1

5

[7]

Q19.

(a) (i) det **W** = 0

B1

Transformed shapes have zero volume

Or equivalent statement
ft volume statements

B1 **2**

(ii) Char eqn is $\lambda^3 - 4\lambda^2 + 4\lambda = 0$

One A mark for each coefficient (not the λ^3)

M1A3

Solving the cubic eqn

M1

$\lambda = 0, 2, (2)$

A1 **6**

Alternative:

Det ($\mathbf{W} - \lambda\mathbf{I}$) = $\begin{vmatrix} 3-\lambda & -1 & 1 \\ 2 & -\lambda & 2 \\ -1 & 1 & 1-\lambda \end{vmatrix}$

$= \begin{vmatrix} 2-\lambda & 0 & 2-\lambda \\ 2 & -\lambda & 2 \\ -1 & 1 & 1-\lambda \end{vmatrix}$

Use of R/C ops. $R_1' \to R_1 + R_3$

M1

$= (2-\lambda)\begin{vmatrix} 1 & 0 & 1 \\ 2 & -\lambda & 2 \\ -1 & 1 & 1-\lambda \end{vmatrix}$

Factor of $(2 - \lambda)$ *correctly extracted*

A1

$= (2-\lambda)\begin{vmatrix} 1 & 0 & 0 \\ 2 & -\lambda & 0 \\ -1 & 1 & 2-\lambda \end{vmatrix}$

$C_1' \to C_3 - C_1$

$= (2-\lambda)^2\begin{vmatrix} 1 & 0 & 0 \\ 2 & -\lambda & 0 \\ -1 & 1 & 1 \end{vmatrix}$

A1A1

$= (2 - \lambda)^2 (-\lambda)$

Complete factorisation attempt

M1

giving eigenvalues 0 and 2 (twice)

A1

(6)

(b) (i) $x - y + z = 0$

B1

1

(ii) $$\begin{bmatrix} 3 & -1 & 1 \\ 2 & 0 & 2 \\ -1 & 1 & 1 \end{bmatrix}\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 3a - b + c \\ 2a + 2c \\ -a + b + c \end{bmatrix}$$

M1A1

$x' - y' + z' = 3a - b + c - 2a - 2c - a + b + c$

M1

$= 0 \Rightarrow P'$ in H also

Shown carefully

A1

4

[13]

Q20.

Expanding fully: $\Delta = x^3 + y^3 + z^3 - 3xyz$

B1

Using row/column operations:
eg $R_1' = R_1 + (R_2 + R_3)$

M1

$$\Rightarrow \Delta = (x + y + z)\begin{vmatrix} 1 & 1 & 1 \\ z & x & y \\ y & z & x \end{vmatrix}$$

With conclusion that $(x + y + z)$ is a factor of the required expression when $k = 3$

A1

NB Any line of argument that leads correctly from $(x + y + z)\ f(x, y, z)$ to $x^3 + y^3 + z^3 - 3xyz$ scores full marks

[3]

Q21.

(a)

$$\Delta = \begin{vmatrix} a-b & b & c \\ b-a & c+a & a+b \\ c(b-a) & ca & ab \end{vmatrix}$$

$C_1' = C_1 - C_2$

M1

$$= (a-b)\begin{vmatrix} 1 & b & c \\ -1 & c+a & a+b \\ -c & ca & ab \end{vmatrix}$$

A1

2

Or Setting $b = a \Rightarrow C_1 = C_2 \Rightarrow \Delta = 0$

Factor theorem

(M1)

$\Rightarrow (a - b)$ a factor of Δ

(A1)

(2)

Or $\Delta = (a - b)(c^3 + a^2b + ab^2 - abc - b^2c - a^2c)$

(M1)

Must be completely correct

(A1)

(2)

(b)

$$= (a-b)\begin{vmatrix} 1 & b-c & c \\ -1 & c-b & a+b \\ -c & a(c-b) & ab \end{vmatrix}$$

$C_2' = C_2 - C_3$

M1

$$= (a-b)(b-c)\begin{vmatrix} 1 & 1 & c \\ -1 & -1 & a+b \\ -c & -a & ab \end{vmatrix}$$

2^{nd} linear factor extracted

A1

e.g. $\Delta = (a-b)(b-c)\begin{vmatrix} 0 & 0 & a+b+c \\ -1 & -1 & a+b \\ -c & -a & ab \end{vmatrix}$

Genuine attempt at both remaining linear factors: e.g. $R_1' = R_1 + R_2$

M1

and then expanding final det.

3rd factor

A1

$\Delta = -(a+b+c)(a-b)(b-c)(c-a)$

All correct

A1

5

Or By cyclic symmetry,
$(b-c)$ and $(c-a)$ are also factors

(M1)
(A1)
(A1)

Final linear factor & checking sign of a coefficient.

(M1)
(A1)

(5)

Or Expanding the determinant fully
$\Delta =$

(M1)
(A1)

Multiplying out
$(a-b)(b-c)(c-a)(a+b+c)$

No fudging, or jumping straight to the answer allowed

(M1)

=

(A1)

Fully correct working to show the two things are identically equal & checking for sign

(A1)

(5)

[7]

Q22.

(a) (i) det $\mathbf{P} = 4a + 6 + 4 + a = 5a + 10$

M1A1

2

(ii) When $a = 3$, det $\mathbf{P} = 25$

ft

B1F

1

(iii) Setting their det $\mathbf{P} = 0 \Rightarrow a = -2$

M1

ft

A1F

2

(b) (i) $\mathbf{P}^{-1} = \frac{1}{25}\mathbf{Q}$

B1

1

(ii) $(\mathbf{PQ})^{-1} = (25\,\mathbf{I})^{-1} = \frac{1}{25}\mathbf{I}$

M1A1

2

Or $(\mathbf{PQ})^{-1} = \mathbf{Q}^{-1}\,\mathbf{P}^{-1}$

Ignore $(\mathbf{PQ})^{-1} = \mathbf{P}^{-1}\,\mathbf{Q}^{-1}$ *if they can make it work*

(M1)

$= \mathbf{Q}^{-1} \cdot \frac{1}{25}\mathbf{Q} = \frac{1}{25}\mathbf{I}$

(A1)

(2)

(iii) det $\mathbf{PQ}$ = det $(25\,\mathbf{I}) = 25^3$ or 15625

M1A1

det $\mathbf{PQ}$ = det $\mathbf{P}$. det $\mathbf{Q}$

Used

M1

$\Rightarrow$ $25^3 = 25 \det \mathbf{Q}$
$\Rightarrow$ $\det \mathbf{Q} = 25^2$ or 625

A1 **4** **[12]**

Q23.

$$\Delta = \begin{vmatrix} y - x & x & x+y-1 \\ x - y & y & 1 \\ y - x & x+1 & 2 \end{vmatrix}$$

attempt at first linear factor, eg $C_1' = C_1 - C_2$

M1

$$= (y-x)\begin{vmatrix} 1 & x & x+y-1 \\ -1 & y & 1 \\ 1 & x+1 & 2 \end{vmatrix}$$

for 1st linear factor (Ignore remaining det.)

A1

$$\Delta = (y-x)\begin{vmatrix} 0 & x+y & x+y \\ -1 & y & 1 \\ 1 & x+1 & 2 \end{vmatrix}$$

Attempt at second linear factor, eg $R_1' = R_1 + R_2$

M1

$$= (y-x)(y+x)\begin{vmatrix} 0 & 1 & 1 \\ -1 & y & 1 \\ 1 & x+1 & 2 \end{vmatrix}$$

For 2nd linear factor (Ignore remaining det.)

A1

Full expansion

M1

$\Delta = (y - x)(y + x)(2 - x - y)$

Completely correct solution

A1 **6**

Or

Setting $y = x \Rightarrow C_1 = C_2 \Rightarrow \Delta = 0$

Factor theorem

(M1)

$\Rightarrow$ $(y - x)$ a factor of Δ

(A1)

Setting $y = -x$ $\Rightarrow$ $R_1 = R_2$

(M1)

So that $R_1' = R_1 + R_2$ $\Rightarrow$ $R_1' = 0$
$\Rightarrow$ $\Delta = 0$ and $(y + x)$ a factor of Δ

(A1)

Genuine attempt at 3rd factor

(M1)

Completely correct solution

(A1)

(6)

Additional notes:
M0 for full expansion from the start with no successful factorisation progress
M1 A1 M0 M1 A0 for full expansion after one factor found and remaining quardric factor left unfactorised (or incorrectly done)
4 + M0 for two correct linear factors but final det incorrectly expanded.
5 + A0 for minor sign error, but correct otherwise

[6]

Q24.

(a) Attempt at either determinant

M1

det **P** = $k - 2$ det **Q** = $3k - 28$

A1A1

3

(b) Use of det (**PQ**) = (det **P**) (det **Q**)

M1

Creating a quadratic

From $(k - 2)(3k - 28) = 16$

dM1

$3k^2 - 34k + 40 = 0$

A1√

$k = \frac{4}{3}$ or 10

CAO

B1

4

Alternative:

$\mathbf{PQ} = \begin{bmatrix} 28 & 2k+15 & 3 \\ 7k+2 & 2k+4 & 2k+2 \\ 48 & 3k+26 & 6 \end{bmatrix}$

B1

Det (**PQ**) = $3k^2 - 34k + 56$

B1$\checkmark$

Equating to 16 and solving

M1

$k = \frac{4}{3}$ or 10

CAO

A1

4

[7]

Q25.

(a) det **A** = 1

B1

1

(b) The z-axis (i.e. $x = y = 0$)

B1

1

(c) Rotation

M1

about the z-axis

ft axis or correct

A1ft

through $\cos^{-1} 0.28$

73.7° (awrt 74°) or 1.29 rads;

$\sin^{-1} 0.96$, $\tan^{-1} \frac{24}{7}$

A1

3

[5]

Q26.

$$\Delta = \begin{vmatrix} 1 & a^2 & bc \\ 0 & b^2 - a^2 & c(a-b) \\ 0 & c^2 - a^2 & b(a-c) \end{vmatrix}$$

Row/column operations (≥ 1 set)
By $R_2' = R_2 - R_1$ and $R_3' = R_3 - R_1$

M1 A1

$$= (b-a)(c-a)\begin{vmatrix} 1 & a^2 & bc \\ 0 & a+b & -c \\ 0 & c+a & -b \end{vmatrix}$$

Taking out factors (≥ 2 attempted)

M1 A1

$$= (b-a)(c-a)(c-b)\begin{vmatrix} 1 & a^2 & bc \\ 0 & a+b & -c \\ 0 & 1 & 1 \end{vmatrix}$$

By $R_3' = R_3 - R_2$ and then expanding fully
(including attempt at all four factors)

M1

$= (a-b)(b-c)(c-a)(a+b+c)$
Or by use of factor theorem and cyclic symmetry

Penalise "global" sign errors at the end only

A1

6

[6]

Q27.

(a) $\begin{vmatrix} 1 & 3 & -1 \\ 2 & k & 1 \\ 3 & 5 & k-2 \end{vmatrix}$

Attempted expansion

M1

$= k^2 - 2k - 10 + 9 + 3k - 5 - 6(k-2) = k^2 - 5k + 6$

AG

A1

2

(b) (i) When $k = 1$, $\Delta \neq 0$ $\Rightarrow$ $\exists$ **one** solution.

Give for correct single soln.

(− 16, 13.5, 14.5) found

B1

(ii) When $k = 2$, $\Delta = 0$ (no unique solution.)

System is $x + 3y - z = 10$

$2x + 2y + z = -4$

$3x + 5y = 6$

B1

$R_1 + R_2 = R_3$ on *both* sides

M1

$\Rightarrow$ $\exists$ ∞-ly many solutions

A1

(iii) When $k = 3$, $\Delta = 0$ (no unique solution.)

System is $x + 3y - z = 10$

$2x + 3y + z = -4$

$3x + 5y + z = 7$

B1

(1) + **(2)** $\Rightarrow$ $3x + 6y = 6$

Or $x + 2y = 2$

(1) + **(3)** $\Rightarrow$ $4x + 8y = 17$

Or $x + 2y = 4\frac{1}{4}$ *etc.*

M1

$\Rightarrow$ System inconsistent and $\nexists$ **no** solutions

A1

7

(c) $\underline{k = 1}$: the (single) point of intersection of 3 planes

B1√

$\underline{k = 2}$: 3 planes meet in a line (or form a sheaf)

B1√

$\underline{k = 3}$: 3 planes form a "prism" (or have three parallel lines of intersection; or have no common intersection)

B1√

ft one of each type (provided matched up correctly)

3

[12]

Q28.

(a) Det **M** = –15 + 12 + 0 – (–12 + 0 – 30)

M1

= 39

A1

2

(b) (i) $V(S_1) = 12 \times 39 = 468$ cm^3

ft

M1 A1

2

(ii) $V(S_2) = 12 \times 39 \times \left(\frac{1}{3}\right)^2 = 52$ cm^3

ft

M1 A1

2

[6]

Q29.

(a) $\Delta = \begin{vmatrix} a+b+c & a+b+c & a+b+c \\ b+c & c+a & a+b \\ b-c & c-a & a-b \end{vmatrix}$

By $R_1' = R_1 + R_2$

M1A1

$= (a+b+c)\begin{vmatrix} 1 & 1 & 1 \\ b+c & c+a & a+b \\ b-c & c-a & a-b \end{vmatrix}$

Factoring out correctly

A1

EXAM PAPERS PRACTICE

or $\begin{vmatrix} a+b+c & b & c \\ 2(a+b+c) & c+a & a+b \\ 0 & c-a & a-b \end{vmatrix}$

By $C_1' = C_1 + C_2 + C_3$

$= (a + b + c) \begin{vmatrix} 1 & b & c \\ 2 & c+a & a+b \\ 0 & c-a & a-b \end{vmatrix}$

Etc.

Method for obtaining remaining factor

M1

$\Delta = 2(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

2 can be anywhere

A1

5

(b) Nothing $a = a$, $b = 3$, $c = 1$

B1

Setting $\Delta = 0$

M1

i.e. $0 = 2(a + 4)(a^2 - 4a + 7) \Rightarrow a = -4$

A1

Showing $(a - 2)^2 + 3 \neq 0$ so only one value of a

Ignore incorrect quadratic factors until here

CSO

M1
A1

5

Alt.

$\begin{vmatrix} a & 3 & 1 \\ 4 & 1+a & a+3 \\ 2 & 1-a & a-3 \end{vmatrix} = 2a^3 - 18a + 56$

(B1)

Equating to zero + solving attempt

(M1)

$2(a + 4)(a^2 - 4a + 7) = 0 \Rightarrow a = -4$

(A1)

$(a - 2)^2 + 3 > 0 \Rightarrow$ no other real solutions.
Or discriminant < 0

(M1A1)

(5)

[10]

Examiner reports

Q2.

A number of students seemed not to know the result det $A^{-1} = \frac{1}{\det A}$ even though this is listed in the specification as one of the results to be learnt. However, in part (b), almost all students were able to show their understanding of the formula det AB = det A × det B . A number of students also did not appreciate the link between the determinant and the area scale factor.

Q3.

The most successful method in part (a) was to find the determinant and set it to equal zero. The least successful approach was to try to combine equations in a more traditional manner. Students simply got themselves lost in the algebra and could not complete the method. The row operation method of elimination was rarely seen but often proved successful and helped with part (b). Part (b) illustrated again that students are not confident or clear about consistent and inconsistent equations readily getting them mixed up.

Q4.

There was a clear factor, $x^2 + y^2 + z^2$, in the third row. Many students failed to find this factor and then struggled through line after line of, often incorrect, algebra. The key method here is to use row and column operations to be able to find and extract factors. The best solutions are clear and state which row or column operations are used. When correct operations were used and $y - x$ was extracted as a factor from $y^2 - x^2 - y + x$, a common error was to give $2(y + x)$ as the other factor as opposed to the correct factor of $y + x - 1$. Those who correctly answered part (a) generally achieved some success on part (b); they could explain why the linear factors could not be zero but did not generally explain why the quadratic factor was not zero.

Q5.

The vast majority of students correctly set and expanded the determinant | $\mathbf{M} - \lambda\mathbf{I}$ | = 0 although the request to show that $\lambda = 2$ was an eigenvector was sometimes overlooked. Part (b) was well done, although a minority of students mixed up methods and tried to solve ($\mathbf{M} - 2\mathbf{I}$)$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$. In part (c)(i), most students were able to easily verify that $\lambda = 4$ was an eigenvector, although a significant number tried to replicate the method from part (a). Part (c)(ii) proved successful for the majority of students although not all of them realised how the answer could have been obtained from 2 × 4 = 8.

Q6.

A good response was received to this question. In part (a) it is not sufficient to give the determinant as 3 and deduce that the matrix is non-singular. The key point to secure the mark is to state that 3 ≠ 0 and then make the deduction. Inverse matrices are well understood, marks were dropped here for the standard types of errors with signs or miscalculation of minor determinants. Part (c) proved more differentiating. A minority of students were not aware of the result $(AB)^{-1} = B^{-1}A^{-1}$ and hence set off on a trail to find

B^{-1} by other, often invalid, means.

Q7.

This was another generally popular and successful question for candidates. Most attempts scored all 4 marks on part (a) and most of those for part (b). However, the approaches taken often provided further evidence that the candidature, in general, was not at the appropriate level of mathematical maturity required for taking their MFP4 paper at this time. Having supplied three equations with a common coefficient of 1 for z, easing the work needed to establish the inconsistency of the system, most candidates preferred to work with the x's and y's instead, which just took a little longer and led more often to the introduction of numerical errors. Also, it was often the case that inconsistency was "established" on the basis that two constants weren't equal, when comparing a couple of equations, but candidates' lack of care in the arithmetic frequently meant that one or other of these two constants wasn't the correct one, which invalidated their conclusion and lost them a mark.

Q8.

The algebraic manipulation of determinants has always been found tricky by at least half of all students in each MFP4 paper set thus far, and this trend continued this summer also.
Part of the difficulty here is in getting students not to expand prematurely; some students just cannot seem to resist doing so immediately, and many others expanded immediately after completing part (a), usually correctly. Giving them one factor, and requiring that they show it is a factor, not only gives them a bit of a start but forces them to do the manipulation. It was disappointing to see how many students, having successfully managed the first factor, lacked the confidence to repeat the process with their own choice of a second. Leaving the expansion until the last possible moment is the only way, for many, to get the complete factorisation correct.

Q9.

Apart from the few who did not seem to know what a transpose matrix was, this was a popular and high-scoring question, even for those with relatively low scores on the paper as a whole. The only really disturbing aspect of the work on display was that in part (b)(ii) so few students seemed to have any idea how to go about finding a linear factor of a fairly simple cubic and then factorising to find the accompanying quadratic factor, which is C1 work. Most students who gained the answer $k = -3$ appeared to get it from a calculator, and very few indeed could use it to find the required quadratic factor; fewer still were able to reason convincingly that this quadratic factor had no real roots, which again is C1 work.

In the final part of the question, many students failed to realise that k was now being replaced in the given matrix **A** by $k - 7$, so that all they had to do was to add 7 to their previous value of k for two straightforward marks. Almost all students who correctly found the answer for this new k did so by going back, working out a cubic characteristic equation (with some large numbers for coefficients), and using a calculator to solve it for them. This was, somewhat reluctantly, awarded the marks if the answer was correct. Most of those students who did not get the correct answer also used this method, but any incorrect working prevented their calculator helping them out and so they gained no marks.

Q10.

This was a straightforward starter to the paper, and was generally found to be so by candidates. However, almost half of all candidates failed to score all 4 marks, due largely to arithmetical slips and carelessness with signs. This was mostly due to unhelpfully lengthy work on row- and/or column–operations after the $(x + y + z)$ factor had been extracted, when the simplest approach was to just go ahead and expand the remaining determinant. This highlights the view that many of these candidates had had insufficient practice with handling determinants algebraically in order to arrive at the point when they might have a "feel" for what is a good approach to take at different stages. Moreover, even in this example, it was about 50-50 whether candidates added R_3 into R_2 or vice versa; the latter approach clearly producing a simpler determinant to continue to work with afterwards, if some thought had been given to it.

Q11.

Most candidates scored highly on this question. In part (b), almost all candidates used the product of the two determinants found in part (a), but a substantial number showed a lack of maturity by expanding the product into a cubic expression and then attempting to factorise it, not always correctly.

Q12.

Most candidates were successful in part (a), though some used more efficient methods than others. Many made a good start in part (b) but were unable to express $2n^2 + 2n + 1$ as the sum of two squares, thus losing the last three marks in the question. Some candidates seemed to be unaware that the 'squares' asked for here had to be the squares of polynomials, or in part (c) the squares of integers.

Q13.

The structure of these questions proved very helpful to candidates, and those who were reasonably careful could generally score at least 16 of this question's 18 marks without any difficulty.

Q14.

As usual, the general treatment of determinants using row/column operations was variably received: those who could, did … and generally scored at least six of the eight marks available. Those who were less clear about the topic generally scored nothing or, at best, scrabbled around for a couple of method marks somewhere. Fortunately, the proportion of candidates in the former camp continues to grow year-by-year.

A major bug-bear for the examiners is the decreasing inclination among candidates to give any hint as to what they are attempting to do with the rows and/or columns, leaving it to the markers to figure it out for themselves. In simple cases such as this, it isn't too much of a problem, but in more difficult cases examiners may be unable to award any marks unless there is a **clear method**.

Q15.

Throughout all the MFP4 papers set thus far, this topic of determinants has proved to be a problem to many candidates. It is clear that while some candidates *have* seen how to perform row and column operations on determinants, others equally clearly have not.

The phrase "expand the determinant" in part (a) was supposed to direct candidates away from using row and/or column operations, but many did not take the hint. Conversely, the set-up in part (b) obviously requires row/column operations to begin with but, once a linear factor has been extracted (the one given), candidates should have been able to spot that they can then expand directly, as they only have a quadratic expression to deal with.

Some candidates went on to use row/column operations very concisely and successfully, but they were very much the exception rather than the rule.

Q16.

Most candidates made a good attempt at the first two parts and only a few got really 'bogged down' in part (c). It was necessary to spotting that **det** (**MN**) = **det** (**M**) **det** (**N**), and these final two marks of the paper were nicely discriminating of the more able candidates' flexibility. Quite a few other candidates managed to see what was going on but didn't quite realise how to explain why the result arose or to identify correctly the x, y and z referred to.

Q17.

It was very pleasing to see a large numbers of candidates solving part (a)'s system of equations using some fairly high-powered techniques (see the end of the marking scheme for these: Cramer's Rule, the augmented matrix method, and the inverse matrix method) to find the unique solution, but most of these are actually more complicated than what was required. The "low-tech" algebraic approach does the job so easily that the others (whilst eminently worth teaching) often turn out to be longer and more prone to errors.

One approach that did turn up for part (b)(ii) is that one can simply substitute the values of x, y and z found in part (a) into the 3rd equation – using the value of a from part (b)(i) – in order to find the value of b required for the final equation to be consistent. This is because the third plane must share the line of intersection of the first two planes, and hence the point on it found in part (a). Only a handful of students actually employed this approach.

Q18.

Almost all candidates did very well at finding the inverse matrix using the 'transposed matrix of co-factors' approach, and only a small proportion forgot the alternating signs and/or the transposition.

Q19.

Incorporating a 3 × 3 matrix inevitably leads to a higher degree of difficulty in finding the characteristic equation, and this frequently proved to be so for many candidates. Once again, however, it was the introductory remark required to explain the significance of a zero determinant that caused most difficulty. Surprisingly few candidates seemed entirely sure about it, and the most popular suggestion was that det **W** = 0 meant that volumes were unchanged. However, the majority of responses dwelt exclusively on the theme of areas, which would have been suitable only had **W** been a 2 × 2 matrix. A very few candidates produced a surprising answer, explaining that **W** was a "dimension-destroying (or crushing)" matrix. This gains no credit, unless candidates then go on to explain that 3-d shapes become 2-d ones, or possibly even lines or points, at this stage. A lot of candidates wrote incoherent and meaningless statements: vague statements such as "it is co-planar" don't carry much weight and certainly don't get the marks.

Q20.

Because of its slightly unusual nature, this short question was put last, so that candidates would not spend time they could better employ elsewhere in trying to figure out what to do with it. In the event, it proved far less of a problem than anticipated. The intention was that candidates would expand the determinant fully to gain the expression $x^3 + y^3 + z^3 - 3xyz$ and then use row/column operations to extract the factor of $(x + y + z)$. Most did indeed do so. The mark most commonly lost was the final one of tying the two ends together, which many failed to do, and examiners were quite strict in not giving full marks for solutions which never got round to joining up two otherwise disparate bits of working.

Q21.

This was the type of question previously handled very badly, but this time most candidates appeared well-drilled in how to manipulate determinants. The greatest obstacles to a completely successful conclusion to the question lay in the widespread instinct to expand the remaining determinant at too early a stage, leaving many candidates unable to factorise a quadratic expression in two variables, for instance; and in the more minor error of assuming that the final expression was fully cyclically symmetric in a, b, c, but failing to check that there was a 'minus sign' difference between what they were expecting and what they had been given.

Q22.

This was the last question on the paper because, since it involves a 3 × 3 matrix, it was potentially the most time-consuming question. However, candidates found the structure of the question very helpful.

Q23.

The manipulation of determinants continues to be a problem area for many candidates. Expanding from the outset is not really a good idea at all, and all who did so failed to cope satisfactorily with the resulting cubic expression. Even amongst those who found one linear factor before expanding, it was almost invariably the case that they were unable to cope with the resulting quadratic factor. This is very disappointing in its own right, especially when they often had a difference of two squares expression clearly written out in front of them.

Many candidates who used the expected row/column operations approach seemed ill-inclined to state anywhere what operations they were doing. This leaves the markers the task of guessing or deciphering the intended approach, and marks are not credited if the working proves too obscure to figure out.

Q24.

This was another straightforward question, and almost all candidates scored highly on it. Apart from a very few who multiplied to get **PQ** first, most were happy to use the result det **PQ** = (det **P**)(det **Q**) to get the required quadratic and find the two values of k.

Q25.

Most candidates answered this question very well, making intelligent use of the results given in the formula booklet. Once again, however, there was a significant proportion of candidates who gave an inappropriate answer, often $z = 0$, when asked for the invariant line in part (b).

Q26.

This was one of the questions for which many candidates seemed unprepared. Many showed no ability in manipulating determinants by using row or column operations. The number of candidates attempting to factorise the determinant by the factor theorem was very small. A few candidates gave the correct answer with little or no working, and these were given the benefit of the doubt. However, it should be noted that a direct approach to factorising determinants becomes less feasible as the degree of the terms increases. Only very capable students are likely to be able to use such methods successfully, although it is encouraging and impressive to see them do so.

Q27.

This was the worst answered question on the paper. Although seven marks were allocated to part (b), many candidates simply wrote down answers without justification. Few seemed to have a correct notion of the connection between the determinant found in part (a) and the number of solutions of the system. A curious mistake was to find values for x, y and z and then to conclude that there were three solutions to the system of equations. A small number of candidates, even though not answering part (b) correctly, were aware of the three possible answers and gave one of each. It was then possible for them to deduce the correct geometrical interpretations for each case and obtain the marks for part (c).

Q28.

Most candidates knew what to do.

Q29.

This question was less well done than Question 7 for the most part. Many candidates knew how to expand a determinant, but were unable to manipulate one using row/column operations. With part (a) proving to be difficult, part (b) was usually even more difficult. Candidates often failed to use part (a) to answer part (b). Very few candidates produced a second quadratic factor for the determinant and only a minority of these justified that the required value of a was unique.